

HiEn: Hierarchical ensemble learning for semi-supervised medical image segmentation

Long Ma, Jie Chen, Nan An, Xinwei Xue^{*}, Chengpei Xu, Weimin Wang, Yi Wang

Dalian University of Technology, Dalian, China

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ABSTRACT

Medical image segmentation has seen growing interest in developing advanced learning paradigms, driven by scarce expert annotations and heavy reliance on domain expertise. However, existing methods frequently encounter performance bottlenecks due to insufficient modeling of cross-model (student and teacher) parameter interactions and inadequate cross-domain knowledge integration between labeled and unlabeled data domains. To this end, this work proposes a hierarchical ensemble learning framework for semi-supervised medical image segmentation. We devise a hierarchical modeling paradigm with dual-teacher constraints that captures complex mapping relationships between student and teacher models. Then we derive a unified computational framework that synergistically optimizes supervised and unsupervised learning objectives, implementing an alternating parameter update strategy for end-to-end training. Extensive experiments across surgical scenarios (*e.g.*, Fetoscopy, Endoscopy) demonstrate that our method achieves superior performance compared to other cutting-edge works. Our code is available at <https://github.com/jch3rn/HiEn>.

1. Introduction

Medical image segmentation delineates structures, tissues, or lesions at the pixel level, enabling precise diagnostics and treatment. Deep learning advances have enhanced segmentation through autonomous feature extraction, robust handling of complex image structures, and reduced noise sensitivity. However, despite their high accuracy [1–3], these methods require extensive paired training data—a challenge given the labor-intensive and costly nature of annotating complex medical images. Consequently, researchers are developing strategies for high-precision segmentation with limited labeled data.

Existing semi-supervised methods [4–6] employ various constraints to exploit unlabeled data, mitigating the scarcity of labeled samples. Some approaches [7,8] use consistency constraints by inputting images of the same object from different perspectives or modalities, thereby ensuring uniform segmentation results. These methods also harness spatial relationships between adjacent pixels to promote spatial coherence. Other techniques [9,10] incorporate uncertainty constraints through probabilistic models to estimate the segmentation distribution for each pixel or region, or by combining predictions from multiple models via ensemble learning for more reliable outcomes. However, these strategies often fail to effectively integrate supervised and unsupervised learning, resulting in suboptimal extraction of unlabeled data and limited network

generalization. Furthermore, coarse training processes and overlapping constraints further impede segmentation performance [11].

To address these challenges, we present HiEn, a hierarchical ensemble learning framework for semi-supervised medical image segmentation. To overcome the confirmation bias and the limited diversity of supervisory signals inherent in standard single-teacher models, HiEn introduces a complementary dual-teacher architecture. By maintaining two teachers with asymmetric update mechanisms, our framework captures diverse knowledge representations. This dual-teacher design serves as a robust structural foundation, enabling more reliable consistency constraints and adversarial feature perturbations. As illustrated in Fig. 1(b), our HiEn implements an interactive parameter propagation mechanism derived from a general hierarchical modeling presented in Fig. 1(a). Quantitative comparisons in Fig. 1(c) demonstrate HiEn's consistent superiority over contemporary methods. Our contributions are rigorously summarized as follows:

- **Dual-Teacher Constrained Hierarchical Modeling:** We establish a new dual-teacher constrained hierarchical modeling mechanism that systematically captures latent correspondences between student and teacher models.
- **Hierarchical Ensemble Learning:** We develop a unified alternating learning framework that synergizes supervised

^{*} Corresponding author.

Email address: xinweixue@dlut.edu.cn (X. Xue).

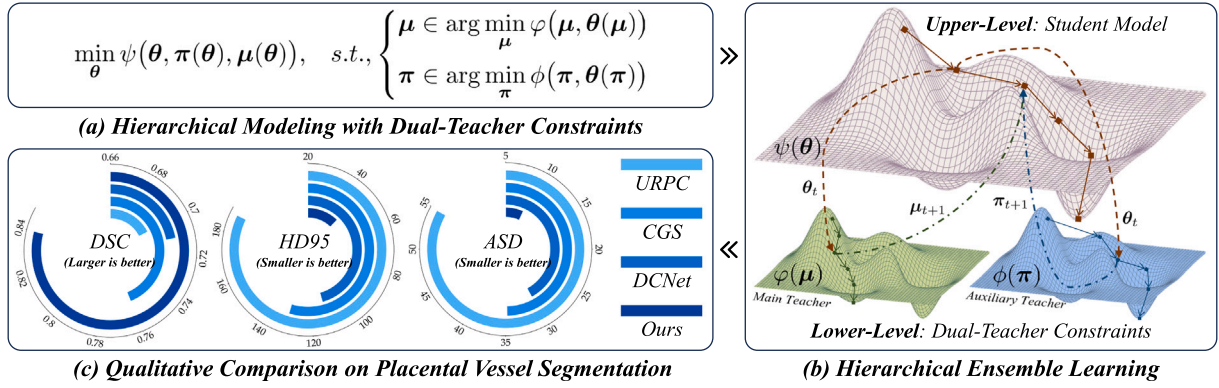


Fig. 1. A Peek into Algorithm of HiEn. Sub-figure (a) illustrates our proposed modeling process for semi-supervised segmentation, where the target student model (θ) serves as the upper-level objective, while the source model is enforced through dual-teacher constraints (μ and π). Sub-figure (b) depicts the hierarchical parameter propagation. Finally, sub-figure (c) showcases performance superiority of our method.

and unsupervised objectives to realize the effective knowledge ensemble with fast and stable convergence.

- **Comprehensive Experimental Validation:** We conduct extensive experiments on challenging surgical imaging benchmarks, including fetoscopic placental vessel segmentation and endoscopic scene segmentation.

2. Related work

2.1. Semi-supervised learning

Semi-supervised learning (SSL) has become a cornerstone for tackling the annotation bottleneck in medical image segmentation by effectively leveraging abundant unlabeled data. The field is largely dominated by two interconnected paradigms: consistency regularization and pseudo-labeling.

Consistency Regularization is founded on the smoothness assumption, which enforces that a model's output should be invariant to small, semantically meaningless perturbations applied to its input or its internal state [12,13]. This principle is operationalized through various perturbation strategies. A highly successful model-level strategy involves a teacher-student architecture, where a student network is trained to produce predictions consistent with a teacher network. The teacher's parameters are typically an exponential moving average (EMA) of the student's, providing a more stable and robust target [14]. This foundational Mean Teacher model has been extended in numerous ways, for example, by incorporating uncertainty estimation to focus consistency losses on more reliable regions [9,15] or by developing sophisticated strategies to select ambiguous targets for regularization [16]. An alternative model-level approach is co-training, which enhances model diversity by training two or more networks with different initializations or architectures. These networks then provide mutual supervision for each other, a technique often referred to as cross pseudo supervision [17–19]. To enforce diversity with greater parameter efficiency, some works employ a single shared encoder with multiple, structurally distinct decoders [20]. Data-level perturbations are also widely used, most notably through weak-to-strong augmentation frameworks where high-confidence predictions from a weakly perturbed image are used as targets for a strongly perturbed counterpart [21,22]. These augmentation pipelines for segmentation are often enhanced with advanced strategies like CutMix or copy-paste techniques to create more challenging and diverse training examples [23–25].

Pseudo-Labeling, also known as self-training, follows an intuitive iterative pipeline: a model is first trained on the available labeled data, then used to predict labels for the unlabeled data. These predictions are then filtered and used as “pseudo-labels” to retrain the model alongside the original ground truth data [26]. While conceptually simple, the

primary pitfall of this approach is confirmation bias, where the model can become overconfident in its own initial, and often incorrect, predictions. This can lead to a feedback loop of error accumulation that degrades performance [27]. Consequently, a vast body of research is dedicated to improving the quality and utility of pseudo-labels. The most direct line of defense is selective labeling, where only pseudo-labels exceeding a certain confidence threshold are utilized. To improve upon simple static thresholds, some methods introduce adaptive or dynamic thresholding strategies that evolve during training [28,29]. A more sophisticated variant is uncertainty-aware selection, which leverages metrics like prediction entropy or Monte Carlo dropout variance to identify and filter out unreliable pseudo-labels [30,31]. Beyond simply filtering, another class of methods focuses on pseudo-label refinement. For example, in a multi-model setup, the disagreement between peer networks can be used to identify and potentially correct noisy labels, with one network's predictions serving as a refined supervisory signal for another [32]. Curriculum learning approaches also address this issue by introducing pseudo-labels in a structured manner, starting with the easiest (most confident) examples and gradually incorporating more challenging ones as the model improves [33].

Modern SSL frameworks rarely use these paradigms in isolation, instead opting for hybrid approaches that capitalize on their complementary strengths. For example, the cross-supervision in co-training models is a natural form of pseudo-labeling, coupled with an implicit consistency constraint. More recently, contrastive learning has been integrated as a powerful auxiliary task to learn more discriminative pixel-level representations from unlabeled data. This involves encouraging features from the same class to be closer in the embedding space while pushing features from different classes apart, proving particularly effective for refining segmentation at object boundaries [34–36]. Despite these advances, generating robust and diverse supervisory signals from unlabeled data remains a central challenge, driving continuous innovation in the field.

2.2. Semi-supervised medical image segmentation

While the core paradigms from general SSL are foundational, their successful application in medical image segmentation often requires specialized adaptations to address the domain's unique characteristics, such as strong anatomical priors and subtle tissue boundaries. A prominent trend is the integration of geometric or topological constraints to guide the learning process. Rather than relying solely on generic data augmentations, some frameworks incorporate shape-aware losses or auxiliary tasks, such as predicting signed distance maps (SDMs) to enforce smoother and more anatomically plausible segmentation boundaries [37–39]. In a similar vein, other recent approaches have explored structure-aware transformations that preserve image topology,

Algorithm 1 Hierarchical Ensemble Learning.

Require: The labeled (D_{su}) and unlabeled (D_{un}) datasets, other hyper-parameters.

```

1: The initial model parameters  $\theta^0, \mu^0, \pi^0$  by random initialization.
2: while not convergence and  $t \leq T$  do
3:   % The supervised learning phase
4:   for  $k = 0 : K - 1$  do
5:      $\mu_{k+1}^t \leftarrow \mu_k^t - \eta_\mu \nabla_\mu \mathcal{L}_{\text{su}}(\mu_k^t; D_{\text{su}})$ ,
6:      $\theta_{k+1}^t = \beta_\theta \theta_k^t + (1 - \beta_\theta) \mu_{k+1}^t$ .
7:   end for
8:   Let  $\theta^t = \theta_K^t$  and  $\mu^t = \mu_K^t$ .
9:   Initialize the parameters of auxiliary teacher by  $\pi^t = \mu^t$ .
10:  % The unsupervised learning phase
11:  for  $l = 0 : L - 1$  do
12:     $\bar{\mu}_l^t = \beta_\mu \mu_l^t + (1 - \beta_\mu) \theta_l^t$ ,
13:     $\pi_{l+1}^t = \beta_\pi \pi_l^t + (1 - \beta_\pi) \theta_l^t$ ,
14:     $\mu_{l+1}^t \leftarrow \bar{\mu}_l^t - \eta_\mu \partial_\mu \mathcal{L}_{\text{un}}(\theta_l^t, \bar{\mu}_l^t, \pi_{l+1}^t; D_{\text{un}})$ ,
15:     $\theta_{l+1}^t \leftarrow \theta_l^t - \eta_\theta \partial_\theta \mathcal{L}_{\text{un}}(\theta_l^t, \mu_{l+1}^t, \pi_{l+1}^t; D_{\text{un}})$ ,
16:  end for
17:  Let  $\theta^{t+1} = \theta_L^t$  and  $\mu^{t+1} = \mu_L^t$ ;  $t = t + 1$ .
18: end while
19: Return  $\theta^T$ .
```

such as those based on Max-tree representations, to create diverse yet structurally valid training samples [40].

Beyond data-level priors, consistency-based methods are frequently enhanced with domain-specific awareness. A key challenge is the quality of pseudo-labels used as consistency targets. To this end, some methods leverage model uncertainty to focus consistency losses on more reliable prediction regions [9]. More advanced strategies go further by actively using knowledge gleaned from labeled data to rectify or correct potentially noisy pseudo-labels before they are used for supervision [41]. The scope of consistency itself has also expanded; in addition to enforcing consistency across different scales of a prediction pyramid [15], recent works like AdaptFRCNet have explored enforcing consistency at a multi-granularity region level and even in the frequency domain, which is adept at capturing features of medical lesions that are subtle in the spatial domain [42].

Furthermore, contrastive learning, when applied to segmentation, is specifically adapted to operate at the pixel- or voxel-level to learn highly discriminative feature representations. This is particularly crucial for distinguishing between similar-looking tissues or precisely delineating irregular lesion boundaries, a common challenge in medical imaging [34,43]. To tackle this directly, some methods propose edge-aware contrastive losses that explicitly use uncertainty maps to guide the sampling of positive and negative pairs from boundary regions, thereby compelling the model to learn more robust edge representations [44]. Collectively, these adaptations illustrate a clear trajectory in the field: a shift from applying generic SSL techniques to developing specialized, hybrid frameworks that integrate anatomical priors and advanced representation learning to overcome the specific hurdles of medical image analysis.

3. Methodology

This section introduces a hierarchical modeling, details the hierarchical ensemble learning mechanism, and finally formulates the training losses and details the network architecture.

3.1. Hierarchical modeling with dual-teacher constraints

The teacher model, trained on labeled data, encapsulates effective task-specific knowledge, which can serve as a valuable resource for the target model trained on unlabeled data. This suggests the presence of a complex interdependent relationship between the source and target

models. Additionally, in light of the consistency learning challenges [20, 21] associated with unlabeled data for student model (denoted as $\mathcal{F}$ with parameters θ), we introduce dual-teacher models (the main teacher $\mathcal{G}$ with parameters μ and the auxiliary teacher $\mathcal{H}$ with parameters π) as constraints. Inspired by the concept of bilevel optimization in hyper-parameter tuning [45], we formulate this complex learning process as a hierarchical dual-model constrained paradigm. Consequently, we define the following idealized hierarchical objective:

$$\min_{\theta} \psi(\theta, \pi(\theta), \mu(\theta)), \quad s.t., \begin{cases} \mu \in \arg \min_{\mu} \varphi(\mu, \theta(\mu)), \\ \pi \in \arg \min_{\pi} \phi(\pi, \theta(\pi)), \end{cases} \quad (1)$$

where θ represents the parameters of student model within the energy function $\psi(\cdot)$. The parameters μ and π denote the parameters of main teacher and auxiliary teacher models within the energy functions $\varphi(\cdot)$ and $\phi(\cdot)$, respectively.

As illustrated in Fig. 1(b), it presents a clear schematic depicting the hierarchical parameter propagation.

3.2. Hierarchical ensemble learning

Eq. (1) effectively captures the ideal theoretical relationship between the student and teacher models. However, computing the exact local optimum ($\arg \min$) for the lower-level constraints at each iteration is often computationally prohibitive for high-dimensional deep networks. To bridge the gap between this theoretical formulation and practical implementation, we develop Algorithm 1 as a feasible heuristic optimization scheme. Inspired by recent successful practices in cross-domain optimization [46,47], we substitute the exhaustive inner-loop optimization with an alternating single-step gradient update combined with an Exponential Moving Average (EMA) momentum. This effectively relaxes the strict hierarchical constraints into a stable, heuristic learning strategy, optimizing the coupled dual-model system without requiring strict stationarity at every iteration. The overall process is presented in Fig. 2, which encompasses both the supervised and unsupervised learning phases.

Supervised Learning. To maintain robustness within the labeled image feature space, we propose the following updated strategy:

$$\begin{cases} \mu_{k+1}^t \leftarrow \mu_k^t - \eta_\mu \nabla_\mu \mathcal{L}_{\text{su}}(\mu_k^t; D_{\text{su}}), \\ \theta_{k+1}^t = \beta_\theta \theta_k^t + (1 - \beta_\theta) \mu_{k+1}^t, \end{cases} \quad (2)$$

where the loss function $\mathcal{L}_{\text{su}}$ represents the supervised loss (see Section 3.3 for details) on labeled data. The symbol D_{su} represents the annotated data for supervised learning. The parameter η_μ is the step-size coefficient. The parameter β_θ is the positive weighting coefficient from the Exponential Moving Average (EMA) strategy [14]. The superscript t and subscript k denote the iteration number for supervised learning phase and the overall computation, respectively.

Unsupervised Learning. To strengthen the coupling property between the source and target model, we first update the dual-teacher model using the EMA strategy to ensure the learning stability, formulated as

$$\begin{cases} \bar{\mu}_l^t = \beta_\mu \mu_l^t + (1 - \beta_\mu) \theta_l^t, \\ \pi_{l+1}^t = \beta_\pi \pi_l^t + (1 - \beta_\pi) \theta_l^t, \end{cases} \quad (3)$$

where β_μ and β_π are the positive weighting coefficients similar to β_θ . The subscript l denotes the iteration number for unsupervised learning phase.

Further, we define the following update process for the main teacher and student model by using the unsupervised loss.

$$\begin{cases} \mu_{l+1}^t \leftarrow \bar{\mu}_l^t - \eta_\mu \partial_\mu \mathcal{L}_{\text{un}}(\theta_l^t, \bar{\mu}_l^t, \pi_{l+1}^t; D_{\text{un}}), \\ \theta_{l+1}^t \leftarrow \theta_l^t - \eta_\theta \partial_\theta \mathcal{L}_{\text{un}}(\theta_l^t, \mu_{l+1}^t, \pi_{l+1}^t; D_{\text{un}}), \end{cases} \quad (4)$$

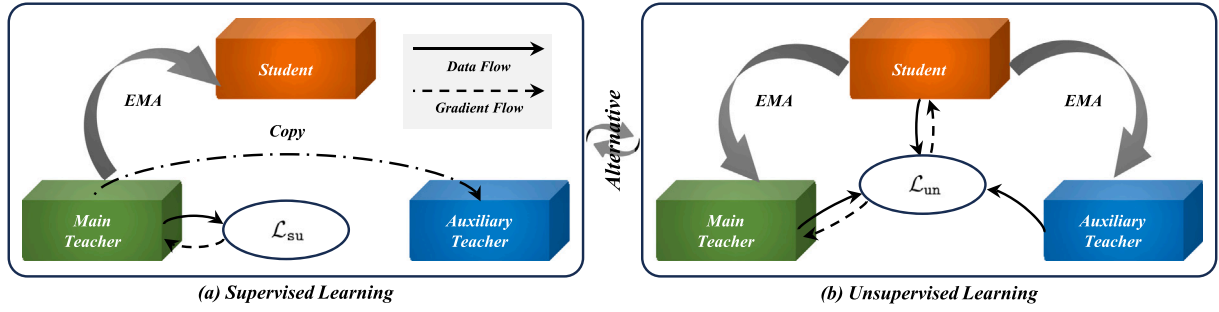


Fig. 2. Computational Workflow Decomposition of HiEn. We emphasize that our primary focus lies in developing an effective learning strategy; as such, we adopt the classical U-Net (see Section 3.4) architecture for segmentation to specifically isolate the benefits of our methodological contributions from architectural complexities.

where η_μ and η_θ are the step-size coefficients. The loss function $\mathcal{L}_{un}$ represents the unsupervised loss (see Section 3.3 for details) on labeled data. The symbol $\mathcal{D}_{un}$ represents the unlabeled data for unsupervised learning.

As detailed in Algorithm 1, the student model's parameters are updated solely via gradient backpropagation during unsupervised training, thereby assuming the risk of performance degradation in that phase. Subsequent refinements incorporate synchronized parameter transfer from the teacher model, enhancing efficiency and numerical stability when integrating supervised signals.

Feature perturbation enhances segmentation model robustness against adversarial attacks. While Virtual Adversarial Training (VAT) [48] injects gradient-based perturbations to maximize prediction-pseudo-label divergence in semi-supervised learning, we extend PSMT's feature perturbation strategy [8] through two key modifications: (1) enforcing stronger feature consistency constraints in T-VAT with multi-level features, and (2) replacing standard data augmentation with input-space randomized noise injection. The adversarial feature perturbation τ is derived from teacher model ensemble responses via:

$$\max_{\tau} f_{KL} \left(\text{Softmax} \left(0.5 \times [\mathcal{G}_{dec}(z) + H_{dec}(z)] \right), \text{Softmax} \left(0.5 \times [\mathcal{G}_{dec}(z_\tau) + H_{dec}(z_\tau)] \right) \right), \quad (5)$$

where $z \triangleq [x_0, x_1, x_2, x_3, x_4]$ concatenates the input x_0 and the intermediate outputs of the student encoder layers (x_i is the i -th layer output, $i = 1, 2, 3, 4$). Here, $z_\tau \triangleq [x_0, x_1, x_2, x_3, x_4 + \tau]$ perturbs the final layer output x_4 by a vector τ , $\mathcal{G}_{dec}$ and H_{dec} are decoders of the teacher and auxiliary teacher models, and f_{KL} aggregates pixel-wise KL divergence between the original and perturbed predictions after $\text{Softmax}(\cdot)$. $\|\tau\|_2 \leq \epsilon$ bounds the perturbation magnitude.

3.3. Training loss

HiEn employs two groups of loss functions comprising both supervised and unsupervised components, as previously outlined in the methodology section.

Supervised Loss. As for supervised learning phase, we define the supervised loss function for updating parameters, formulated as:

$$\mathcal{L}_{su} = \lambda_1 \mathcal{L}_{dice}(\mu; \mathcal{D}_{su}) + \lambda_2 \mathcal{L}_{ce}(\mu; \mathcal{D}_{su}), \quad (6)$$

where $\mathcal{L}_{dice}$ represents the Dice loss [49]. The loss $\mathcal{L}_{ce}$ represents the Cross-Entropy (CE) loss [50]. The coefficients λ_1 and λ_2 are positive hyperparameters used to control the trade-off between the two loss terms. The CE loss emphasizes pixel-wise classification accuracy, while the Dice loss focuses on spatial consistency.

Unsupervised Loss. Inspired by VAT [48], we enforce multi-level consistency using input perturbations and feature-space constraints. For input perturbations, we utilize weak-strong augmentation pairs [17,26]:

auxiliary teacher models process weakly augmented inputs (e.g., flips), while both student and teacher models receive strongly augmented inputs (e.g., blur). The feature consistency loss $\mathcal{L}_{fc}$ aligns feature representations from student $\mathcal{F}$ and teachers $\mathcal{G}/\mathcal{H}$:

$$\mathcal{L}_{fc} = -\log \frac{\sum_{\mathcal{K} \in \{\mathcal{G}, \mathcal{H}\}} e^{\cos(\bar{\mathcal{F}}^m, \bar{\mathcal{K}}^m)/\sigma}}{\sum_{\mathcal{K} \in \{\mathcal{G}, \mathcal{H}\}} e^{\cos(\bar{\mathcal{F}}^m, \bar{\mathcal{K}}^m)/\sigma} + \sum_{\mathcal{K} \in \{\mathcal{F}, \mathcal{G}, \mathcal{H}\}} e^{\cos(\bar{\mathcal{F}}_{su}^m, \bar{\mathcal{K}}^m)/\sigma}}, \quad (7)$$

where $\cos(\cdot)$ denotes cosine similarity, and $\bar{\mathcal{K}}^m$ represents that the feature is processed after an MLP projection head. The variable $\sigma = 0.1$ is the temperature coefficient. The value m indicates the m -th network layer (m is empirically set as 4). $\bar{\mathcal{F}}_{su}$ represents that its input comes from the data with label.

The segmentation consistency loss $\mathcal{L}_{sc}$ enforces prediction agreement between different models within the constraint of ℓ_2 loss $\mathcal{L}_{mse}$, formulated as

$$\mathcal{L}_{sc} = \mathcal{L}_{mse}(\mathcal{F}(\theta), \mathcal{G}(\mu)) + \mathcal{L}_{mse}(\mathcal{F}(\theta), \mathcal{H}(\pi)) + \mathcal{L}_{mse}(\mathcal{G}(\mu), \mathcal{H}(\pi)). \quad (8)$$

The total unsupervised loss is concluded as:

$$\mathcal{L}_{un} = \mathcal{L}_{sc}(\theta, \mu, \pi; \mathcal{D}_{un}) + \mathcal{L}_{fc}(\theta, \mu, \pi; \mathcal{D}_{un}). \quad (9)$$

3.4. Network architecture

Our framework is built upon a U-Net-based encoder-decoder architecture, specifically adapted for semi-supervised learning.

Encoder. We employ a ResNet-34 model [51], pre-trained on the ImageNet dataset, as the primary feature extractor. This choice leverages its robust feature initialization and inherent residual connections, which facilitate smoother gradient flow during training. The encoder's initial convolutional layer is modified to handle various input channel configurations, making the architecture flexible for diverse medical imaging modalities. As an input image passes through the encoder, it is progressively downsampled to learn a rich hierarchy of abstract representations.

Decoder. For precise pixel-level prediction, the decoder symmetrically reconstructs the spatial resolution. At each stage, feature maps are upsampled via 2×2 transposed convolutions and then concatenated with corresponding high-resolution feature maps from the encoder through long-range skip connections. This core U-Net mechanism is vital, as it fuses deep, semantic context with fine-grained spatial information, enabling accurate localization of object boundaries.

Output Heads. The network culminates in two parallel outputs. The primary head uses a final 1×1 convolutional layer to map the refined, full-resolution representation from the decoder to the desired class logits. Concurrently, a projection head, inspired by practices in representation learning [52], processes bottleneck features from the encoder. This head uses global average pooling and a Multi-Layer Perceptron (MLP) to project high-dimensional features into a compact, 128-dimensional embedding, which is then exclusively used to calculate the feature consistency loss.

Table 1
Quantitative results on fetoscopic placental vessel segmentation.

Class	Metric	Full	MT	CPS	CCT	ICT	UAMT	DCT	URPC	DCNet	CGS	Ours
Blood Vessel	DSC↑	0.57	0.57	0.57	0.50	0.57	0.60	<u>0.61</u>	0.59	0.52	0.56	0.71
	HD95↓	111.3	116.8	118.2	131.3	110.0	118.6	<u>103.6</u>	106.0	146.1	117.4	86.9
	ASD↓	15.47	16.02	18.74	23.26	19.53	14.90	<u>14.67</u>	17.86	20.33	18.95	12.86
Ablation Tool	DSC↑	0.48	0.65	0.62	0.54	0.57	0.66	0.64	0.60	0.73	<u>0.80</u>	0.84
	HD95↓	112.5	66.8	68.7	88.8	96.4	68.1	70.5	81.9	84.5	<u>43.6</u>	36.9
	ASD↓	77.69	30.23	20.72	39.79	46.42	31.87	39.04	39.34	43.90	9.53	<u>10.61</u>
Fetus	DSC↑	0.92	0.94	0.83	0.86	0.90	0.92	0.91	0.90	0.88	0.92	0.97
	HD95↓	81.7	177.7	193.5	110.4	94.2	126.4	172.0	375.2	108.8	233.2	11.5
	ASD↓	36.17	49.86	82.12	36.54	32.70	47.80	65.10	107.51	<u>26.38</u>	75.90	3.89
Avg.	DSC↑	0.66	0.72	0.67	0.63	0.68	0.73	0.72	0.70	0.71	<u>0.76</u>	0.84
	HD95↓	101.8	120.4	126.8	110.2	<u>100.2</u>	104.4	115.4	187.7	113.1	131.4	45.1
	ASD↓	43.11	32.04	40.53	33.20	32.89	31.52	39.61	54.90	<u>30.20</u>	34.79	9.12

4. Experiments

4.1. Datasets

FetReg dataset. The FetReg dataset [53] for fetoscopic placental vessel segmentation comprises four semantic classes: background, placental vessel, ablation tool, and fetus. Due to limited computational resources, we curated a subset of high-quality videos (Video001, 003, 008) with optimal visibility and minimal artifacts, yielding 379 annotated frames and 1182 unannotated frames. Video014 (110 annotated frames) and Video020 (123 annotated frames) were used as the validation and test sets, respectively.

Endoscapes dataset. The Endoscapes dataset [54], specifically its public Endoscapes-Seg50 subset, provides a benchmark for surgical scene segmentation. It consists of frames from laparoscopic cholecystectomy videos, annotated with seven distinct anatomical labels. For our semi-supervised experiments, we augmented this labeled data with unlabeled frames from the unreleased Endoscapes-Seg201 collection, while adhering to the official data splits. This dataset is particularly challenging due to the high complexity of surgical scenes, which frequently feature instrument-tissue interactions and occlusions.

LIDC-IDRI dataset. The LIDC-IDRI dataset [55] is a standard benchmark for lung nodule segmentation from CT scans. Our experiments specifically target the segmentation of clinically challenging Ground-Glass Nodules (GGNs), which are notable for their ambiguous and ill-defined margins. The full dataset was partitioned into a training set of 992 slices and an independent test set of 46 slices. To evaluate performance under our standard data-scarce condition, the training set was configured with a 1/4 labeled ratio, resulting in 248 labeled and 744 unlabeled slices for training. To further test the model's robustness under extreme scarcity, we also evaluated performance using an even more stringent 1/8 labeled ratio (124 labeled, 868 unlabeled slices).

4.2. Implementation details

Compared Methods. We first established a fully supervised baseline (denoted as “Full”), which was trained exclusively on the limited labeled data without utilizing any unlabeled samples. Furthermore, we compared our framework against ten advanced semi-supervised methods, including MT [14], ICT [56], CCT [18], CPS [17], DCT [19], UAMT [9], URPC [15], PSMT [8], RCPS [57], DCNet [58], CGS [59] and DuCiSC [60].

Evaluation Metrics. The Dice Similarity Coefficient (DSC) [61] which measures spatial overlap between predictions and ground truth, indicating volumetric consistency; the 95% Hausdorff Distance (HD95) [62] which assesses boundary accuracy via the 95th percentile of maximum surface distance, minimizing outlier effects; and the Average Surface Distance (ASD) [63] which computes average symmetric surface distances to evaluate topological alignment.

Parameter Settings. We employed SGD with a momentum of 0.9 and a weight decay of 0.0001. The initial learning rate was set to 0.01 and decayed following a poly schedule with a power of 0.9. The total batch size was 4, consisting of 2 labeled and 2 unlabeled images in each mini-batch. We trained the framework for a total of 3×10^4 iterations. For our specific modules, the temperature coefficient σ in the feature consistency loss (Eq. 7) was set to 0.1, and the perturbation bound ϵ for the adversarial feature perturbation (Eq. 5) was set to 2.0. The EMA decay hyper-parameters $[\beta_\theta, \beta_\mu, \beta_\pi]$ were defined as [0.98, 0.95, 0.95].

4.3. Comparison with state-of-the-art methods

Fetoscopic Placental Vessel Segmentation. For fetoscopic placental vessel segmentation, the proposed HiEn framework establishes a new state-of-the-art, outperforming all comparison methods by a significant margin (Table 1). In terms of region overlap, HiEn improves the Dice Similarity Coefficient (DSC) by 10.5% over the strongest competitor, CGS. Crucially, while CGS achieves a competitive DSC, it fails to maintain boundary consistency, yielding a high HD95 of 131.4. In contrast, HiEn demonstrates superior shape preservation by reducing the HD95 to 45.1, which is approximately one-third of the error produced by CGS. This indicates that our framework not only detects vessel regions more accurately but also delineates their boundaries with significantly higher precision than the representative consistency-based and pseudo-labeling approaches. Moreover, the detailed class-wise evaluations integrated into Table 1 further demonstrate that HiEn achieves robust performance across a diverse range of categories, particularly securing top-tier results on the challenging “Blood Vessel” and “Ablation Tool” classes. Furthermore, to evaluate robustness under extreme data scarcity, we conducted an additional experiment using only half of the available labeled data. As presented in Table 3, while reducing the training data naturally leads to a general performance degradation across all methods due to the limited feature representation, HiEn maintains exceptional stability under this setting, demonstrating its ability to mitigate the reliance on extensive annotations.

The qualitative results, shown in Fig. 3, visually confirm that these numerical improvements translate to more complete and coherent vessel segmentations. The visual comparison highlights how HiEn overcomes the typical failure modes of other semi-supervised strategies. For instance, pseudo-labeling-based methods like UAMT can sometimes introduce artifacts or incorrect segmentation boundaries, particularly in ambiguous regions. Conversely, other consistency regularization methods like DCNet may struggle to maintain structural continuity, resulting in fragmented vessel structures. Our HiEn framework, which advances the consistency regularization paradigm, adeptly avoids these issues. It successfully preserves the intricate structural integrity of the vessels while preventing the inclusion of erroneous regions, demonstrating the effectiveness of our hierarchical ensemble approach in producing highly accurate and robust results.

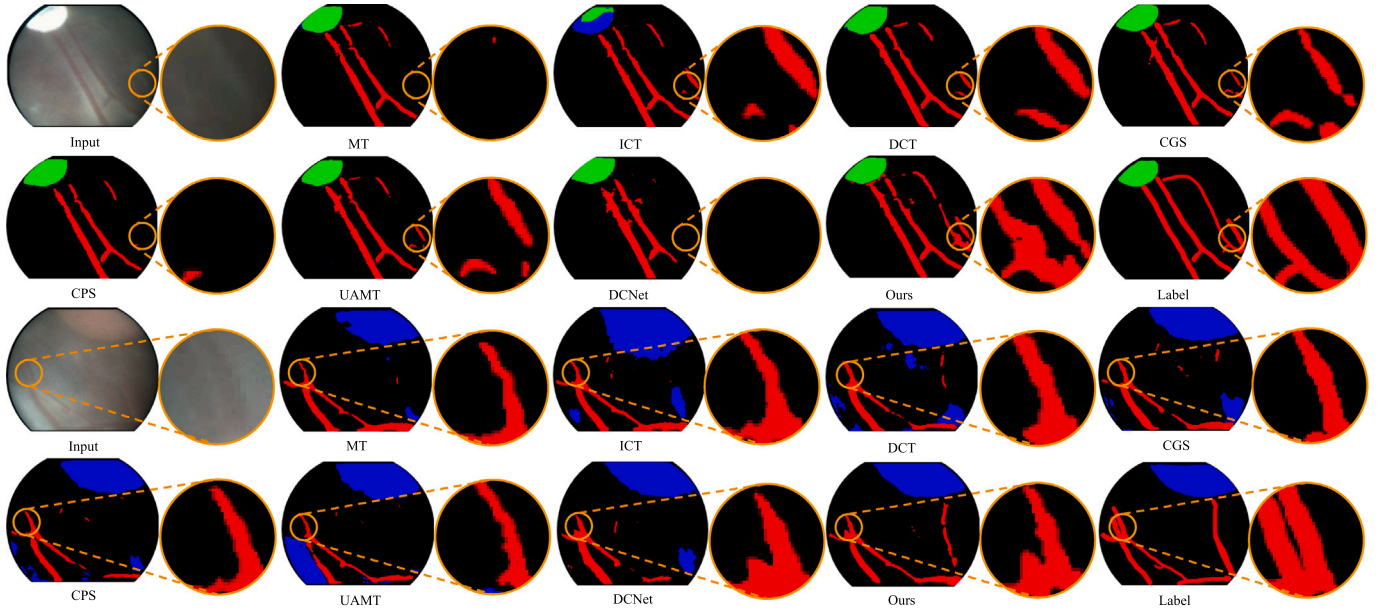


Fig. 3. Visual comparison of fetoscopic placental vessel segmentation. In the segmentation masks, red, green, and blue correspond to blood vessels, fetuses, and surgical tools, respectively.

Table 2
Quantitative results on endoscopic scene segmentation.

Class	Metric	Full	MT	CPS	CCT	ICT	UAMT	DCT	URPC	DCNet	CGS	Ours
Cystic Plate	DSC↑	0.22	0.18	0.25	0.20	0.20	0.19	0.23	0.18	0.15	<u>0.32</u>	0.34
	HD95↓	184.8	228.0	204.0	213.3	200.9	314.0	<u>188.9</u>	299.0	251.4	234.6	190.2
	ASD↓	<u>124.2</u>	160.4	140.6	150.1	145.5	257.1	109.5	251.5	196.9	190.0	140.0
Calot's Triangle	DSC↑	0.28	0.52	0.31	0.26	0.33	0.47	0.33	0.34	<u>0.51</u>	0.37	0.30
	HD95↓	375.8	156.3	228.4	579.2	<u>160.3</u>	291.3	460.1	179.8	167.3	192.9	171.8
	ASD↓	359.1	145.3	207.3	560.3	98.1	282.4	436.5	164.1	<u>145.2</u>	167.5	154.7
Cystic Artery	DSC↑	0.32	0.30	0.31	0.35	0.36	0.35	0.40	0.27	<u>0.39</u>	0.34	0.37
	HD95↓	198.1	193.3	240.4	184.4	171.5	210.8	116.8	330.1	144.0	<u>139.5</u>	149.2
	ASD↓	116.2	126.7	184.4	136.4	114.6	158.1	58.9	291.0	80.2	<u>78.0</u>	98.5
Cystic Duct	DSC↑	0.20	0.31	0.36	0.33	0.35	0.35	0.37	0.33	<u>0.43</u>	0.35	0.48
	HD95↓	348.2	194.7	191.5	267.0	179.5	170.0	168.1	198.2	127.1	207.0	<u>142.7</u>
	ASD↓	296.5	129.3	128.4	221.2	122.3	119.3	112.6	138.6	60.2	150.8	<u>94.6</u>
Gallbladder	DSC↑	0.77	0.56	0.81	0.80	0.79	0.79	0.79	0.81	0.77	<u>0.83</u>	0.83
	HD95↓	188.9	312.9	119.7	166.5	135.4	130.5	169.0	149.6	125.8	<u>110.9</u>	100.9
	ASD↓	71.4	120.1	48.6	62.8	40.9	48.7	65.5	56.0	74.6	<u>44.2</u>	53.3
Tool	DSC↑	0.74	0.62	0.75	0.79	0.79	0.81	0.75	0.79	0.75	0.82	<u>0.81</u>
	HD95↓	106.9	239.8	93.8	101.5	114.2	107.7	122.9	82.7	103.7	95.1	<u>85.8</u>
	ASD↓	25.5	80.9	27.4	39.1	34.5	26.1	43.1	<u>22.1</u>	31.7	27.5	20.5
Avg.	DSC↑	0.42	0.41	0.46	0.45	0.47	0.49	0.48	0.45	<u>0.50</u>	<u>0.50</u>	0.52
	HD95↓	233.8	220.8	179.6	252.0	160.3	204.0	204.3	206.6	<u>153.2</u>	163.3	140.1
	ASD↓	165.5	127.1	122.8	195.0	92.6	148.6	137.7	153.9	98.1	109.7	<u>93.6</u>

Endoscopic Scene Segmentation. For the highly challenging task of endoscopic scene segmentation, our method demonstrates a clear quantitative advantage. As detailed in Table 2, it achieves state-of-the-art performance by securing the top rank for both DSC and HD95, improving upon the second-best method, DCNet, by a notable margin. When compared to ICT, another prominent consistency-based framework, our model shows a compelling trade-off. While ICT narrowly surpasses our model in the ASD metric, our top-tier performance in the primary metrics for both volumetric accuracy (DSC) and critical boundary precision (HD95) underscores our framework's superior overall balance and robustness. This robustness is further evidenced by the class-wise breakdown. Under severe class imbalance, methods like MT or DCT occasionally achieve higher overlap on isolated minor structures such as Calot's triangle or the cystic artery, but they frequently suffer

from severe cross-class instability and catastrophic boundary degradation elsewhere. In contrast, the dual-teacher consistency constraints within HiEn function as a robust global regularizer. By preventing the model from aggressively over-optimizing specific local textures, HiEn trades off these localized peaks to avoid structural collapse across the entire scene, yielding the most balanced and reliable segmentation. Furthermore, when evaluated under the extreme 1/2 labeled setting reported in Table 3, specific competing methods like CCT exhibit an artificial inflation in volumetric overlap but suffer a catastrophic structural collapse with an HD95 reaching 225.3. This anomaly stems from the sequential data split, which inherently possesses lower inter-patient variance and fewer extreme boundary ambiguities. While standard models tend to overfit localized tissue volumes in this lower-variance environment, HiEn's dual-teacher constraints successfully resist such volumetric

Table 3

Quantitative results on fetoscopic placental vessel (Feto) and endoscopic scene (Endo) segmentation under the 1/2 labeled setting.

Task	Metric	Full	MT	CPS	CCT	ICT	UAMT	DCT	URPC	DCNet	CGS	Ours
<i>Feto</i>	DSC↑	0.630	0.643	0.641	0.607	0.649	0.683	0.615	0.621	0.599	<u>0.687</u>	0.701
	HD95↓	231.1	228.7	181.2	227.5	187.6	193.0	250.7	236.1	197.9	<u>171.3</u>	112.6
	ASD↓	112.2	126.6	72.9	109.4	72.6	118.8	162.5	121.6	<u>54.4</u>	62.9	31.2
<i>Endo</i>	DSC↑	0.456	0.474	0.463	0.493	0.456	0.473	0.465	0.470	0.392	0.416	0.476
	HD95↓	243.9	217.8	217.3	225.3	178.2	<u>141.9</u>	223.3	222.1	251.6	242.8	<u>127.9</u>
	ASD↓	189.4	167.1	151.9	155.7	118.3	<u>76.1</u>	169.6	159.6	174.6	180.0	69.2

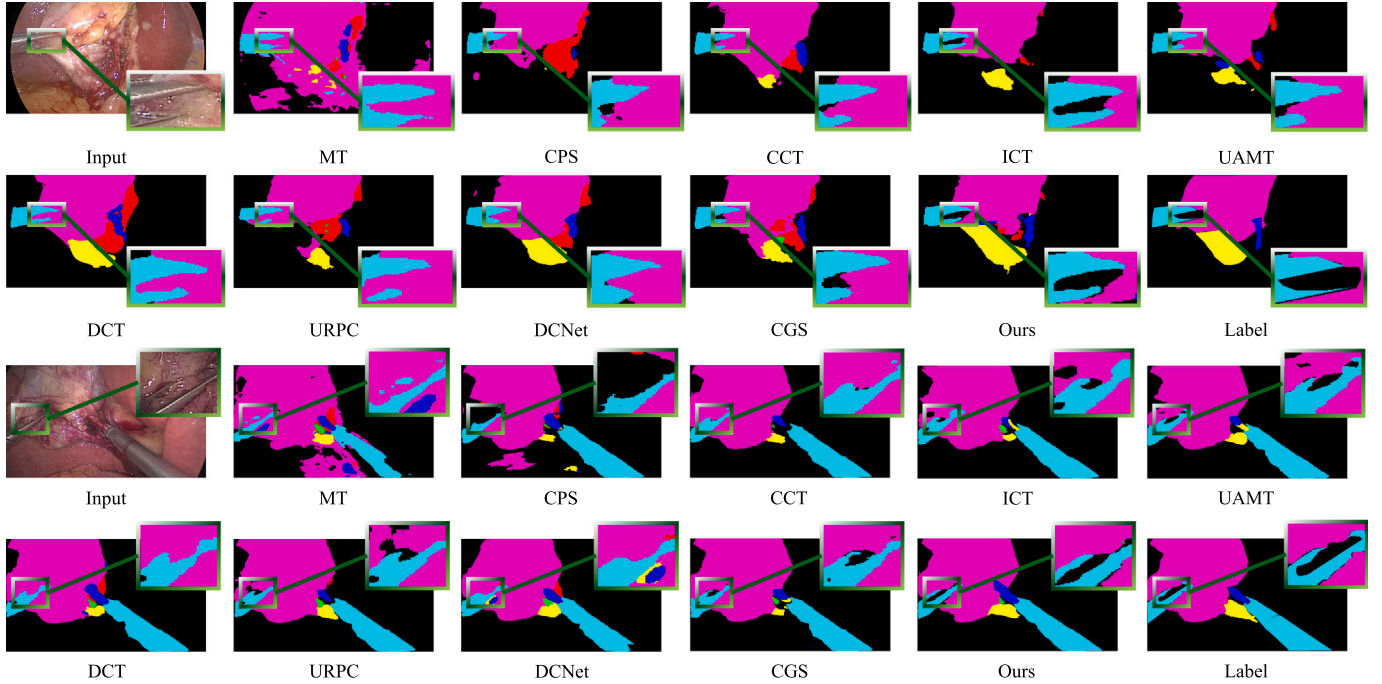


Fig. 4. Visual comparison of endoscopic scene segmentation. The color legend is as follows: Red denotes the Cystic Plate; Green, Calot's Triangle; Blue, the Cystic Artery; Yellow, the Cystic Duct; Magenta, the Gallbladder; and Cyan, the tool.

overfitting. By leveraging the reduced boundary conflicts, HiEn learns cleaner topological structures from the unlabeled data, yielding a distinct improvement in boundary distance metrics despite a natural, slight drop in overall overlap.

This quantitative strength is directly explained by our model's sophisticated and consistent ability to interpret complex instrument-tissue interactions, as vividly illustrated across the two different scenarios in Fig. 4. In the first example (top row), the primary challenge is to avoid misclassifying the background space within the instrument's jaws as gallbladder tissue (magenta). Here, both our framework and ICT prove successful, correctly identifying the background and demonstrating the strength of the consistency-based approach. However, the second example (bottom row) reveals the key differentiator. In this more challenging occlusion, nearly every other method, including ICT, now fails by incorrectly classifying the background space as part of the tool (cyan). Our framework is the only one that remains accurate, correctly perceiving the gaps as background. This ability to maintain precision across multiple, varied, and complex occlusion scenarios is the definitive visual proof of our model's more advanced spatial understanding and is the key to its state-of-the-art performance.

Lung Nodule Segmentation. The lung nodule segmentation task is particularly challenging due to the inherently ambiguous and faint margins of Ground-Glass Nodules (GGNs). Under data-scarce conditions, our proposed approach establishes a new state-of-the-art. As shown in

Table 4, our method consistently outperforms leading techniques under both the 1/4 and the extreme 1/8 labeled settings. Notably, even when the labeled data is reduced to just 1/8 of the training set, our HiEn framework maintains its state-of-the-art performance, achieving a DSC of 0.715. This demonstrates a strong robustness to varying degrees of label scarcity.

These quantitative gains are explained by our model's superior ability to interpret ambiguous visual information, a point vividly illustrated in the magnified view of Fig. 5. The comparison reveals a spectrum of failures in other methods: PSMT and DuCiSC fail completely, producing blank outputs, while MT and CPS struggle significantly with the nodule's scale, creating segmentations that are either too large or too small. UAMT produces a much more competitive result, correctly approximating the nodule's size. However, a closer look reveals that the shape of its segmentation is less faithful to the true anatomical curvature seen in the ground truth. Our model, in contrast, captures both the scale and the precise shape with higher fidelity, resulting in a segmentation that is remarkably close to the expert label. Furthermore, Fig. 6 confirms this high performance is consistently maintained across nodules with diverse appearances, underscoring the model's strong generalization capability.

Computational Overhead. A comprehensive evaluation of computational overhead is presented in Table 5. The analysis reveals that during the training phase, the HiEn framework's VRAM consumption

Table 4
Quantitative results on the lung nodule segmentation.

Ratio	Metric	MT	CPS	CCT	ICT	UAMT	DCT	URPC	PSMT	RCPS	DuCiSC	Ours
1/4	DSC↑	0.655	0.700	0.663	0.627	0.659	0.700	0.701	0.698	0.700	0.717	<u>0.714</u>
	HD95↓	21.56	15.19	17.86	33.80	24.02	<u>14.63</u>	21.01	18.52	15.05	16.86	13.73
	ASD↓	15.10	<u>8.41</u>	12.87	28.35	11.32	11.47	14.46	9.55	10.81	13.64	8.14
1/8	DSC↑	0.629	0.695	0.542	0.685	0.613	0.680	0.516	<u>0.706</u>	0.692	0.705	0.715
	HD95↓	28.60	33.97	42.77	20.03	30.61	15.20	54.04	<u>14.53</u>	28.02	27.32	12.51
	ASD↓	25.62	13.34	33.69	14.92	24.19	12.93	51.67	<u>12.64</u>	22.87	17.07	8.30

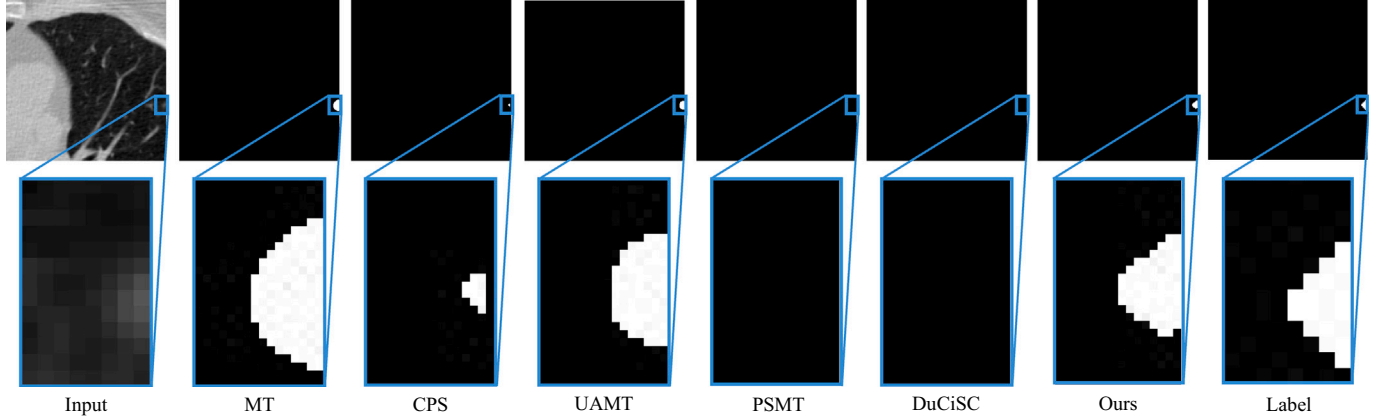


Fig. 5. Magnified visual comparison for lung nodule segmentation. The detailed views highlight the boundary prediction accuracy on a single challenging nodule.

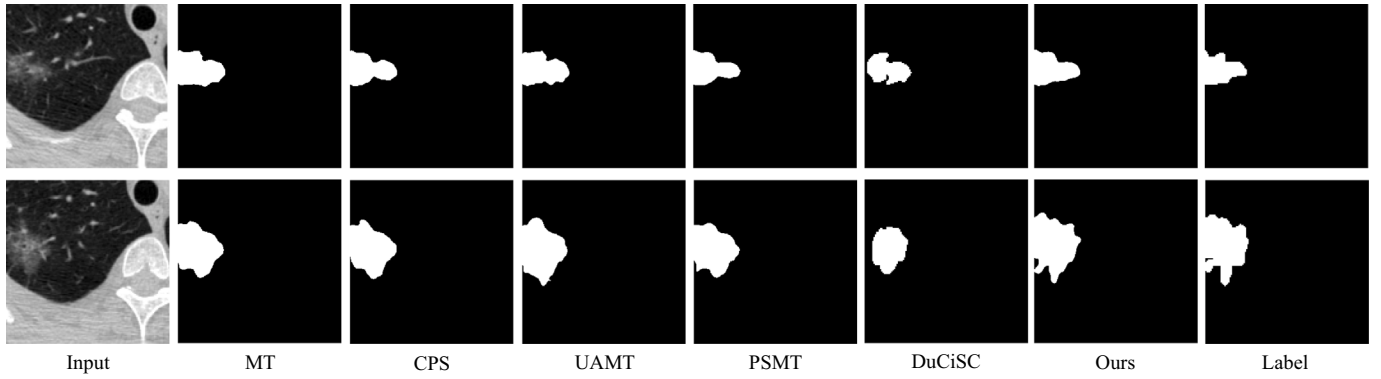


Fig. 6. Visual comparison of lung nodule segmentation on multiple cases. The results showcase the robustness of our method across different nodule appearances.

Table 5
Comprehensive comparison of computational overhead across different methods.

Metric	Full	MT	CPS	CCT	ICT	UAMT	DCT	URPC	DCNet	CGS	Ours
Training Time / Iter. (ms)↓	103.7	138.5	644.5	153.1	170.0	159.3	111.0	120.8	177.4	248.0	218.4
Training VRAM (GB)↓	1.21	1.36	2.29	3.26	1.09	1.56	1.55	1.27	1.02	1.49	2.25
Inference FLOPs (G)↓	27.32	27.32	27.32	80.44	27.32	27.32	27.32	27.36	23.96	16.76	27.32
Inference Speed (FPS)↑	123.1	118.0	126.4	37.7	125.2	126.0	126.6	121.0	106.0	124.7	121.4

and time per iteration are higher than those of standard dual-model baselines like MT and UAMT, yet remain comparable to or lower than other complex architectures such as CCT and CPS.

Regarding inference efficiency, since the teacher models are discarded after training, the final student model introduces no additional theoretical computational overhead. Its complexity, measured in FLOPs, remains identical to several competing methods at 27.32 G. The empirically measured inference speed of 121.35 FPS is also highly competitive and on par with the majority of the other semi-supervised approaches. This indicates that the increased training cost is a one-time, offline

expenditure that does not compromise the model's efficiency during deployment.

4.4. Ablation study

Ablation on Encoder Backbone. To validate our choice of ResNet-34 as the encoder, and demonstrate the plug-and-play applicability of our framework to non-CNN architectures, we conducted an ablation study comparing several alternative backbones. We evaluated a vanilla U-Net, variants using ResNet-18 and ResNet-50, and a Transformer-based Swin-UNet [64]. All models were trained under

Table 6
Quantitative comparison of different encoder backbones.

Encoder	DSC $\uparrow$	HD95 $\downarrow$	ASD $\downarrow$	Training Time
U-Net (Vanilla)	0.7636	113.33	23.82	7h15m
ResNet-18	0.7884	68.98	15.46	8h
ResNet-34 (Ours)	0.8377	45.11	9.12	8h30m
ResNet-50	0.8306	58.66	10.78	14h45m
Swin-UNet	0.7903	55.14	12.60	16h30m

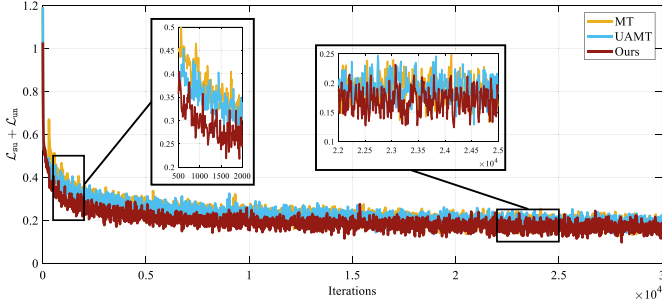


Fig. 7. Convergence curve.

identical experimental conditions. As presented in Table 6, all ResNet-based encoders substantially outperform the vanilla U-Net encoder. ResNet-34 achieves the optimal trade-off. It yields the highest performance across all metrics without the overfitting risk and computational overhead associated with the deeper ResNet-50 model. Interestingly, HiEn successfully accommodates the Swin-UNet backbone and yields a highly competitive DSC of 0.7903. This result surpasses both the vanilla U-Net and ResNet-18. The performance gap between Swin-UNet and ResNet-34 is primarily driven by the data-hungry nature of Vision Transformers under semi-supervised settings [65]. The inherent inductive biases of convolutional architectures remain highly advantageous for delineating fine-grained surgical structures. Nevertheless, the stable convergence of Swin-UNet confirms that our hierarchical dual-teacher modeling is architecture-agnostic and versatile across both CNN and Transformer paradigms.

Convergence Analysis. Our proposed method, HiEn, demonstrates markedly superior convergence properties in terms of both initial speed and long-term stability when compared to baseline approaches like MT and UAMT. As illustrated in Fig. 7, the training loss curve for HiEn exhibits a distinctly steeper and more rapid decline during the initial training phase, signifying that our framework learns fundamental features much more quickly. Following this initial acceleration, HiEn converges to and maintains a lower average loss value, and its oscillations consistently display a smaller amplitude compared to the other methods. In stark contrast, the comparative methods not only

converge to a higher loss but also exhibit more pronounced and erratic fluctuations, indicating training instability likely stemming from conflicting gradient updates in their loosely-coupled objectives. This superior performance is a direct result of our hierarchical framework's principled optimization hierarchy, which mitigates such update conflicts and guides the models along a more stable and efficient learning trajectory.

Impact of Supervised Loss Weights. To determine the optimal form for our supervised loss, we conducted a detailed analysis of the impact of weighting between the Dice and Cross-Entropy (CE) losses, revealing that their interplay is critical for achieving robust and accurate segmentation. Relying solely on $\mathcal{L}_{ce}$ ($\lambda_1 = 0, \lambda_2 = 1$) results in poor boundary delineation due to its primary focus on pixel-wise classification, which leads to fragmented masks and a suboptimal Dice Similarity Coefficient (DSC). Conversely, employing only $\mathcal{L}_{dice}$ ($\lambda_1 = 1, \lambda_2 = 0$) produces more coherent structures by directly optimizing for overlap, but this comes at the cost of degraded boundary accuracy, reflected in a higher Hausdorff Distance (HD95). Our experiments conclusively demonstrate that an equally weighted combination ($\lambda_1 = 0.5, \lambda_2 = 0.5$) achieves the best overall performance; as detailed in the quantitative results in Fig. 8, this balanced configuration yields the highest DSC of 0.8377, along with the lowest HD95 and ASD. The qualitative results presented in Fig. 9 further corroborate these findings, illustrating that the balanced approach generates segmentation outputs that are visually most consistent with the ground truth labels. This highlights the complementary nature of the two functions: $\mathcal{L}_{ce}$ ensures pixel-level precision and contributes to stable training, while $\mathcal{L}_{dice}$ enforces spatial coherence.

Ablation of Different Components. We conducted a comprehensive ablation study to dissect the contribution of each core component within our HiEn framework, with quantitative results shown in Fig. 10. Our baseline model, trained solely with supervised loss ($\mathcal{L}_{su}$), establishes a performance floor. The introduction of the segmentation consistency loss ($\mathcal{L}_{sc}$) provides the most substantial performance leap, improving the DSC by a remarkable 8.85%. Building on this, the feature consistency loss ($\mathcal{L}_{fc}$) further refines the results with an additional 1.93% DSC gain. Finally, integrating our adversarial feature perturbation mechanism (T-VAT) significantly boosts model robustness, contributing another large 5.68% increase in DSC. This cumulative improvement is not merely quantitative; its underlying reason is visually substantiated by the feature maps in Fig. 11.

Ablation on Segmentation Consistency Loss. We adopted the Mean Squared Error (MSE) for our segmentation consistency loss following the established standard practice in semi-supervised learning frameworks such as Mean Teacher. To validate this conventional choice against the specific challenges of medical image segmentation, we compared MSE against Cross-Entropy (CE), JS Divergence, KL Divergence, and L1 Loss. As detailed in Table 7, MSE significantly outperforms all other alternatives on boundary precision metrics. Unbounded losses like CE and KL heavily penalize predictions and force the model to memorize noisy pseudo-boundaries. Conversely, MSE acts as a bounded metric

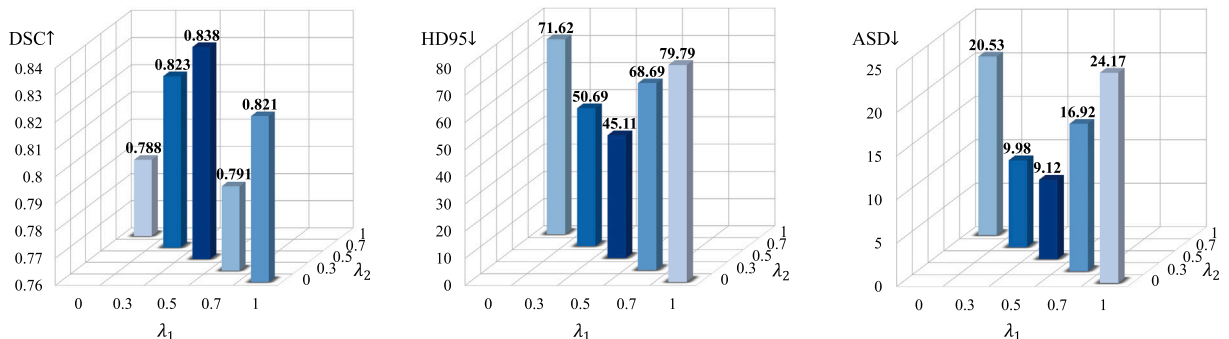


Fig. 8. Quantitative comparison of different loss weights for the supervised loss.

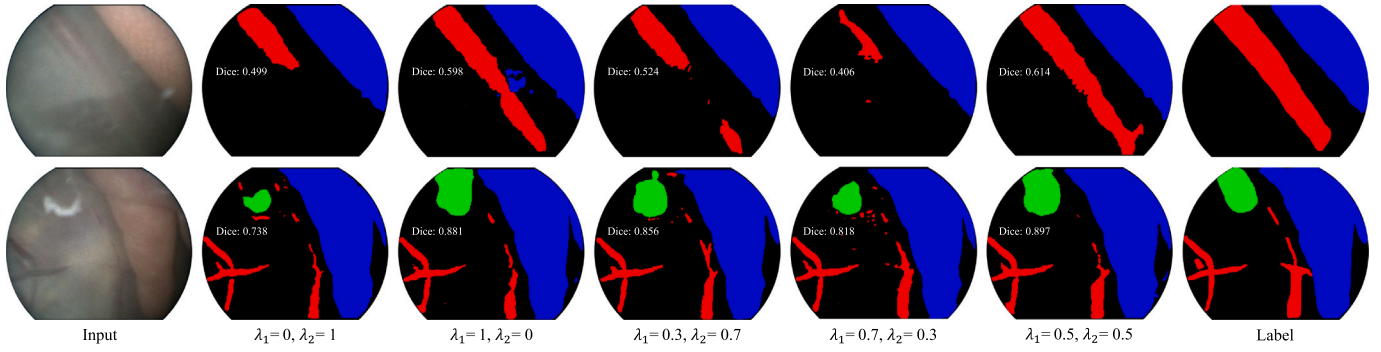


Fig. 9. Qualitative comparison of different loss weights.

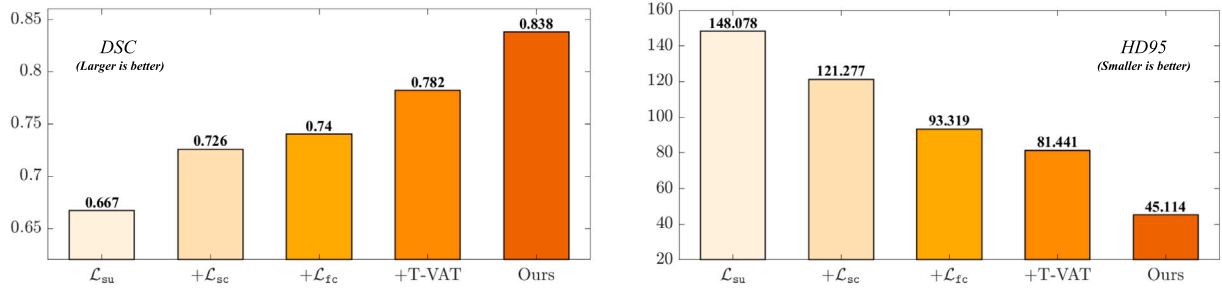


Fig. 10. Quantitative results for the ablation of different components.

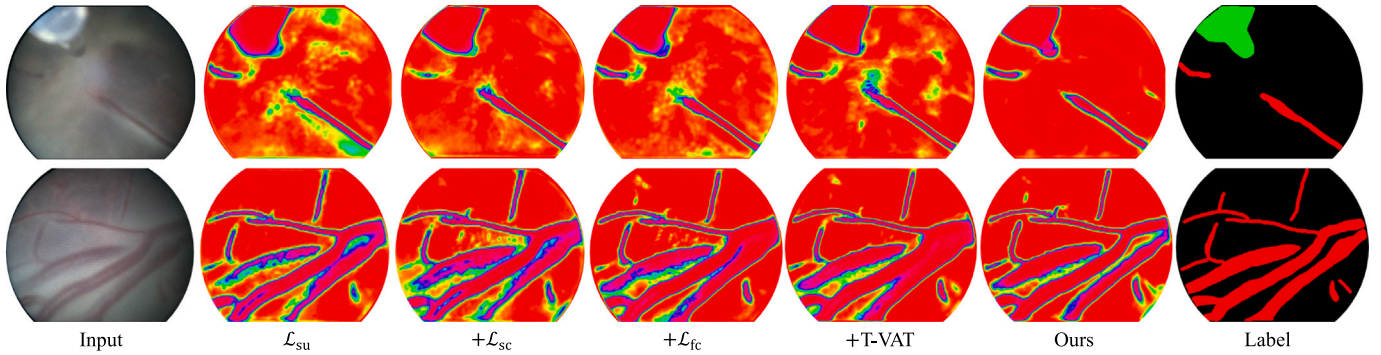


Fig. 11. Visualizing the impact of ablating different components on the model's feature maps. Each map shows how the model's representational focus is refined by the inclusion of each component.

Table 7

Ablation study of different segmentation consistency loss functions ($\mathcal{L}_{sc}$).

Loss Function	DSC $\uparrow$	HD95 $\downarrow$	ASD $\downarrow$
L1 Loss	0.790	65.45	14.54
KL Divergence	0.792	83.94	14.48
JS Divergence	0.796	51.54	10.43
Cross-Entropy (CE)	0.806	65.87	14.98
MSE (Standard)	0.838	45.11	9.12

that is highly tolerant of teacher prediction errors. This bounded nature prevents the amplification of confirmation bias and justifies its widespread adoption as the default consistency metric in the field.

Ablation on Feature Consistency Design. We conducted additional ablations to validate the robustness of our feature consistency loss design. We systematically evaluated different extraction layers and projection head dimensions. As detailed in Table 8, the deepest encoder bottleneck layer ($m = 4$) yields the optimal performance. Shallower

layers inherently contain excessive local spatial details. These premature features easily hinder the global semantic alignment required for cross-model feature matching. Furthermore, the 128-dimensional projection provides the most balanced representation capacity. Smaller dimensions heavily restrict information flow and naturally degrade overall performance. Larger dimensions significantly increase the risk of overfitting on noisy pseudo-labels under limited supervision. These comprehensive results empirically justify our architectural choices and strongly demonstrate the overall robustness of the proposed feature consistency loss.

Effectiveness of Bidirectional Alternating EMA. To validate our parameter update strategy, we compared our bidirectional alternating EMA transfer against a classical unidirectional EMA variant, as shown in Table 9. In the unidirectional setting, the teacher is strictly updated as the EMA of the student across all phases. Evaluated on the FetReg dataset, the unidirectional variant experiences a noticeable performance drop, with the DSC decreasing to 0.82 and the HD95 increasing to 50.1. This indicates that a strictly unidirectional structure is more

Table 8
Ablation study of different feature consistency design choices.

Metric	Extraction Layer m			Projection Dimension		
	$m = 2$	$m = 3$	$m = 4$ (Ours)	64-D	128-D (Ours)	256-D
DSC $\uparrow$	0.770	0.802	0.838	0.806	0.838	0.808
HD95 $\downarrow$	54.02	63.61	45.11	60.51	45.11	51.68
ASD $\downarrow$	16.12	16.34	9.12	11.56	9.12	12.18

Table 9
Ablation study comparing the unidirectional EMA with our bidirectional alternating EMA strategy on the FetReg dataset.

Variant	DSC $\uparrow$	HD95 $\downarrow$	ASD $\downarrow$
Unidirectional EMA	0.818	50.10	10.24
Bidirectional alternating EMA (Ours)	0.838	45.11	9.12

Table 10
Sensitivity analysis of the temperature coefficient σ and the perturbation bound ϵ .

Metric	Temperature Coefficient σ				Perturbation Bound ϵ			
	0.05	0.1 (Ours)	0.2	0.5	0.5	1.0	2.0 (Ours)	4.0
DSC $\uparrow$	0.811	0.838	0.811	0.809	0.785	0.804	0.838	0.788
HD95 $\downarrow$	64.54	45.11	54.13	55.38	60.90	59.93	45.11	59.48
ASD $\downarrow$	14.48	9.12	10.16	10.55	11.23	13.76	9.12	12.08

susceptible to accumulating early errors and amplifying confirmation bias on ambiguous unlabeled data. By contrast, our bidirectional design effectively breaks this error loop. The supervised phase injects accurate knowledge to pull the model back on track, while the unsupervised phase safely aggregates exploratory features. This dynamic successfully prevents gradient divergence and maintains iterative stability, yielding much better structural integrity at complex boundaries.

Hyperparameter Sensitivity Analysis. To validate our choice of key hyperparameters, we conducted a sensitivity analysis on the FetReg dataset for the temperature coefficient σ and the perturbation bound ϵ . The results are presented in Table 10. Our choice of $\sigma = 0.1$ is not only empirically validated but also aligns with common practice in influential contrastive learning frameworks [52]. Similarly, $\epsilon = 2.0$ provides sufficient adversarial robustness without distorting feature representations, a value consistent with the effective range discussed in the original VAT literature [48]. The analysis shows that our framework maintains robust performance across a reasonable range of values, while values that are too small fail to provide meaningful regularization and excessively large values can lead to performance degradation.

5. Limitations

Despite the strong overall performance of HiEn, analyzing its limitations remains important for understanding the boundaries of the

proposed framework. Sub-optimal predictions mainly occur under extreme conditions involving extremely thin structures coupled with low-contrast boundaries. For instance, when segmenting fine-grained placental vessel branches with exceptionally weak intensity gradients, as illustrated in Fig. 12, the model occasionally produces topological discontinuities or fragmented masks. Fundamentally, this vulnerability is intrinsically linked to consistency-based supervision. When teacher models yield highly uncertain predictions in challenging regions, the consistency constraint may inadvertently propagate this uncertainty to the student. Addressing this issue by integrating uncertainty-aware noise filtering and explicit topological priors will be the focus of our future work.

6. Conclusion

In this work, we introduced HiEn, a novel hierarchical ensemble learning framework to address the challenges of semi-supervised medical image segmentation. Our approach formulates a dual-teacher constrained hierarchical modeling paradigm that systematically captures the complex interdependencies between student and teacher models. Through a unified computational framework with an alternating optimization strategy, HiEn effectively synergized supervised and unsupervised learning, ensuring stable and efficient knowledge transfer from limited labeled data. Extensive experiments on diverse surgical imaging benchmarks validated the superiority of our method over a wide range of state-of-the-art techniques. By mitigating the reliance on extensive expert annotations, HiEn provided a powerful and promising solution for developing high-precision automated medical image analysis tools, paving the way for further advancements in the field.

CRedit authorship contribution statement

Long Ma: Methodology, Conceptualization. **Jie Chen:** Writing – original draft, Data curation. **Nan An:** Visualization, Validation. **Xinwei Xue:** Writing – review & editing. **Chengpei Xu:** Formal analysis. **Weimin Wang:** Software. **Yi Wang:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Iterative stability and approximation analysis

This appendix provides a theoretical analysis of Algorithm 1 from two complementary perspectives. We note that the following analysis is intended to provide theoretical intuition and practical justification for the proposed alternating update strategy, rather

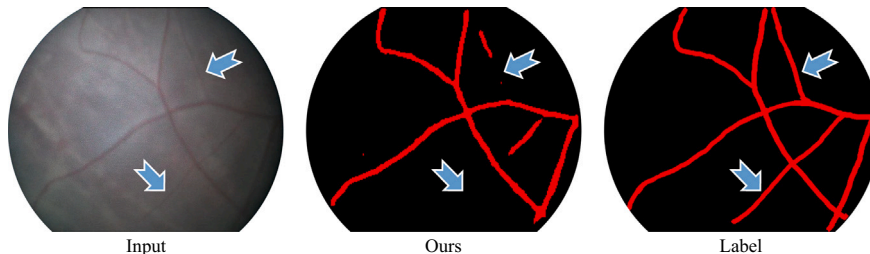


Fig. 12. Visual illustration of a typical failure case.

than rigorous proofs under fully general non-convex conditions. Specifically, we establish the *iterative stability* of the bidirectional alternating EMA updates under standard simplifying assumptions, and characterize the *approximation quality* of Algorithm 1 relative to the idealized hierarchical objective in Eq. (1). The results are consistent with the empirical behavior observed in our experiments and are meant to complement, rather than replace, the empirical validation in the main text.

A.1. Notation and assumptions

Let Θ , $\mathcal{M}$, and Π denote the parameter spaces of the student θ , main teacher μ , and auxiliary teacher π , respectively. We denote by $\mathcal{L}_{su}^*$ the stationary value of the supervised loss. The analysis is conducted under the following standard assumptions.

Assumption 1 (Local Smoothness and PL Condition). The supervised loss $\mathcal{L}_{su}$ is locally L_{su} -smooth and satisfies the Polyak-Łojasiewicz (PL) condition with coefficient $\alpha_{su} > 0$ in a neighborhood of a stationary point μ^* :

$$\frac{1}{2} \|\nabla_{\mu} \mathcal{L}_{su}(\mu)\|^2 \geq \alpha_{su} (\mathcal{L}_{su}(\mu) - \mathcal{L}_{su}^*). \quad (\text{A.1})$$

The unsupervised loss $\mathcal{L}_{un}$ is locally L_{un} -smooth jointly in (θ, μ, π) .

Remark. The PL condition is widely adopted in modern deep learning optimization analysis since it permits linear convergence without requiring global convexity, making it significantly more realistic for deep segmentation models than a global strong convexity assumption.

Assumption 2 (Bounded Stochastic Gradients). There exists a finite constant $G < \infty$ such that $\|\nabla_{\mu} \mathcal{L}_{su}\| \leq G$, $\|\partial_{\theta} \mathcal{L}_{un}\| \leq G$, and $\|\partial_{\mu} \mathcal{L}_{un}\| \leq G$ for all iterates.

Assumption 3 (EMA Coefficients and Step Sizes). The EMA decay coefficients satisfy $\beta_{\theta}, \beta_{\mu}, \beta_{\pi} \in (0, 1)$. The learning rates satisfy $\eta_{\mu} \leq \frac{1}{L_{su}}$ and $\eta_{\theta} \leq \frac{1}{L_{un}}$.

A.2. Coupling stability in the supervised phase

Lemma 1 (Supervised Phase Coupling Stability). Under Assumptions 1–3, after k iterations of the supervised phase, the coupling gap $\Delta_k^{sup} \triangleq \|\theta_k^i - \mu_k^i\|$ satisfies

$$\Delta_k^{sup} \leq \beta_{\theta}^k \Delta_0^{sup} + \frac{\beta_{\theta}(1 - \beta_{\theta}^k)}{1 - \beta_{\theta}} \eta_{\mu} G. \quad (\text{A.2})$$

Proof. From the update rules $\mu_{k+1} = \mu_k - \eta_{\mu} \nabla_{\mu} \mathcal{L}_{su}(\mu_k)$ and $\theta_{k+1} = \beta_{\theta} \theta_k + (1 - \beta_{\theta}) \mu_{k+1}$, we compute the difference directly:

$$\theta_{k+1} - \mu_{k+1} = \beta_{\theta} \theta_k + (1 - \beta_{\theta}) \mu_{k+1} - \mu_{k+1} = \beta_{\theta} (\theta_k - \mu_{k+1}).$$

Taking norms and applying the triangle inequality:

$$\Delta_{k+1}^{sup} = \beta_{\theta} \|\theta_k - \mu_{k+1}\| \leq \beta_{\theta} (\|\theta_k - \mu_k\| + \|\mu_k - \mu_{k+1}\|) \leq \beta_{\theta} \Delta_k^{sup} + \beta_{\theta} \eta_{\mu} G.$$

Unrolling this linear recurrence yields Eq. (A.2). $\square$

Remark. Since $\beta_{\theta} \in (0, 1)$, the transient term $\beta_{\theta}^k \Delta_0^{sup} \rightarrow 0$ as $k \rightarrow \infty$. The coupling gap converges to a bounded steady state:

$$\Delta_{\infty}^{sup} = \frac{\beta_{\theta} \eta_{\mu} G}{1 - \beta_{\theta}}.$$

With the empirical settings $\beta_{\theta} = 0.98$ and a poly-decayed learning rate, this quantity is well-controlled throughout training. Crucially, the coupling gap *cannot diverge* under Assumptions 1–3.

A.3. Boundedness in the unsupervised phase

Lemma 2 (Unsupervised Phase Boundedness). Under Assumptions 1–3, within the unsupervised phase the following bounds hold for all inner iterations l :

$$\|\pi_{l+1}^i - \theta_{l+1}^i\| \leq \beta_{\pi} \|\pi_l^i - \theta_l^i\| + \eta_{\theta} G, \quad (\text{A.3})$$

$$[4\beta\eta] \|\mu_{l+1}^i - \bar{\mu}_l^i\| \leq \eta_{\mu} G. \quad (\text{A.4})$$

Proof. Bound (A.3). From $\pi_{l+1} = \beta_{\pi} \pi_l + (1 - \beta_{\pi}) \theta_l$ and $\theta_{l+1} = \theta_l - \eta_{\theta} \partial_{\theta} \mathcal{L}_{un}$:

$$\pi_{l+1} - \theta_{l+1} = \beta_{\pi} (\pi_l - \theta_l) + \eta_{\theta} \partial_{\theta} \mathcal{L}_{un}.$$

Taking norms and applying Assumption 2 yields Eq. (A.3). Unrolling the recurrence gives the steady-state deviation $\frac{\eta_{\theta} G}{1 - \beta_{\pi}}$, which is finite under Assumption 3.

Bound (A.4). From $\mu_{l+1} = \bar{\mu}_l - \eta_{\mu} \partial_{\mu} \mathcal{L}_{un}$, taking norms directly gives $\|\mu_{l+1} - \bar{\mu}_l\| \leq \eta_{\mu} \|\partial_{\mu} \mathcal{L}_{un}\| \leq \eta_{\mu} G$. $\square$

A.4. Overall stability theorem

Theorem 1 (Uniform Boundedness of Parameter Divergence). Under Assumptions 1–3, the joint parameter trajectory $\{(\theta_k, \mu_k, \pi_k)\}$ generated by Algorithm 1 satisfies the following:

- (i) $\|\theta - \mu\|$ and $\|\theta - \pi\|$ remain uniformly bounded throughout training.
- (ii) After each supervised phase of K iterations, for sufficiently large K , the coupling gap asymptotically satisfies $\|\theta_K^i - \mu_K^i\| \leq C_1$, where $C_1 = \frac{\beta_{\theta} \eta_{\mu} G}{1 - \beta_{\theta}}$ is independent of epoch t .
- (iii) Within each unsupervised phase, $\|\pi_l - \theta_l\|$ converges to a bounded neighborhood of radius $C_2 = \frac{\eta_{\theta} G}{1 - \beta_{\pi}}$.

Proof Sketch. By Lemma 1, each supervised phase contracts $\|\theta - \mu\|$ toward the bounded steady-state radius C_1 , irrespective of the state inherited from the preceding unsupervised phase. For sufficiently large K , the transient term $\beta_{\theta}^K \Delta_0^{sup}$ becomes negligible and the gap is dominated by C_1 . By Lemma 2, the unsupervised phase evolves from this bounded initialization under stable EMA updates and bounded gradient perturbations, maintaining $\|\theta - \pi\|$ within radius C_2 . Since C_1 and C_2 are finite constants independent of the training epoch, the alternating dynamics cannot produce uncontrolled parameter growth. This establishes uniform boundedness of the entire joint trajectory. $\square$

A.5. Bidirectional vs. unidirectional EMA: confirmation bias analysis

The key structural advantage of Algorithm 1 over classical unidirectional EMA lies in *error isolation*. To formalize this, we track the supervised suboptimality at the output of each supervised phase:

$$B_t \triangleq \mathcal{L}_{su}(\mu_K^t) - \mathcal{L}_{su}^*,$$

where μ_K^t denotes the parameters after K supervised gradient steps in epoch t . This definition ensures that B_t reflects the state at the exact point handed off from the supervised phase to the unsupervised phase, making the subsequent contraction bound directly applicable.

In a classical unidirectional EMA scheme, the target is continuously updated as $\mu \leftarrow \beta \mu + (1 - \beta) \theta$. Any prediction error committed by θ on unlabeled data is structurally absorbed into μ , which drives the model away from the supervised optimum. This may cause B_t to grow monotonically under noisy pseudo-labels, creating a positive feedback loop for confirmation bias.

Algorithm 1 structurally interrupts this loop. After the unsupervised phase, the supervised phase resets μ via gradient descent on clean labeled data. Under Assumption 1, the PL condition combined with the standard descent lemma for L -smooth functions guarantees that gradient

descent with step size $\eta_\mu \leq \frac{1}{L_{su}}$ satisfies:

$$B_{t+1} \leq \rho_{su}^K B_t, \quad \rho_{su} = 1 - \eta_\mu \alpha_{su} \in (0, 1). \quad (\text{A.5})$$

This establishes that after each supervised phase of K steps, the supervised phase acts as a *contraction mapping* on the confirmation bias gap B_t with factor $\rho_{su}^K \in (0, 1)$: the suboptimality *cannot amplify across epochs*, regardless of the errors introduced during the unsupervised phase. The empirical evidence in Table 9 of the main text—where the bidirectional design consistently improves DSC and HD95 over the unidirectional variant—is fully consistent with this theoretical guarantee.

A.6. Connection to the hierarchical objective

We now characterize the approximation quality of Algorithm 1 relative to the idealized hierarchical objective in Eq. (1).

Under Assumption 1, gradient descent on $\mathcal{L}_{su}$ with step size $\eta_\mu \leq \frac{1}{L_{su}}$ produces a sequence satisfying:

$$\mathcal{L}_{su}(\mu_K) - \mathcal{L}_{su}^* \leq \rho_{su}^K (\mathcal{L}_{su}(\mu_0) - \mathcal{L}_{su}^*), \quad \rho_{su} = 1 - \eta_\mu \alpha_{su} \in (0, 1). \quad (\text{A.6})$$

This follows directly from the PL condition combined with the standard descent lemma for L -smooth functions. Eq. (A.6) shows that after K supervised steps, μ_K achieves an ϵ -suboptimal solution for the lower-level supervised objective, with the suboptimality shrinking at the linear rate ρ_{su}^K .

Therefore, focusing on the tractable lower-level component of Eq. (1), the optimization error of the teacher update relative to its idealized stationary solution can be quantified by two distinct bounding components:

$$\text{Lower-level Suboptimality: } \mathcal{L}_{su}(\mu_K) - \mathcal{L}_{su}^* \leq \mathcal{O}(\rho_{su}^K), \quad (\text{A.7})$$

$$\text{EMA Coupling Lag: } \|\theta_K - \mu_K\| \leq \mathcal{O}\left(\frac{\eta_\mu G}{1 - \beta_\theta}\right). \quad (\text{A.8})$$

The suboptimality bound in Eq. (A.7) decreases exponentially with the number of supervised phase steps K . The parameter coupling lag in Eq. (A.8) remains tightly controlled under practical learning-rate schedules and can be effectively reduced by decreasing the learning rate during training, as achieved by the poly decay schedule employed in our experiments. Together, these two bounds characterize the approximation quality of the lower-level teacher optimization in Algorithm 1, providing a principled justification for its role as a practically stable realization of the hierarchical dependency structure described in Eq. (1).

Data availability

Data will be made available on request.

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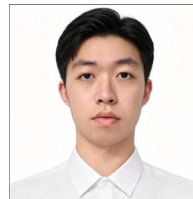
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Author biography



Long Ma received his M.S. degree in Software Engineering at Dalian University of Technology, Dalian, China, in 2019. He completed his Ph.D. degree in Software Engineering at the Dalian University of Technology, Dalian, in 2024. He is currently a Postdoctoral Fellow in the School of Software Technology at Dalian University of Technology. His research interests include low-level and high-level computer vision in adverse imaging conditions.



Jie Chen received the B.E. degree in software engineering from Dalian University of Technology, Dalian, China, in 2024. He is currently pursuing the M.S. degree in software engineering at Dalian University of Technology, Dalian, China. His research interests include medical image analysis and computer vision.



Nan An received the B.S. degree in computer science and technology from Liaoning University, China, in 2021. He is currently pursuing the Ph.D. degree in software engineering at Dalian University of Technology, China. His research interests include computer vision and deep learning.



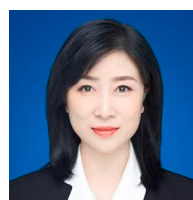
Xinwei Xue received the B.S. and M.S. degrees in the Dept. of Information and Media Sciences from the University of Kitakyushu, Japan, in 2008 and 2010, respectively. She received Ph.D. degree in 2013 from Waseda University, Japan. She is currently an Associate Professor in the School of Software Technology at Dalian University of Technology. Her current research interests include machine learning, image processing and computer vision.



Chengpei Xu received the B.E. degree in systems engineering from the National University of Defense Technology, Changsha, China, in 2014, the master's degree in systems engineering from the University of New South Wales, Sydney, NSW, Australia, in 2018, and the Ph.D. degree in computing sciences from the University of Technology Sydney, Sydney, in 2022. He is currently a Professor in the School of Software Technology at Dalian University of Technology. His research interests include computer vision, scene text detection and recognition, multimodal data fusion, 3-D point cloud processing, and artificial intelligence in eLearning.



Weimin Wang received the B.S. degree from Shanghai Jiao Tong University in 2009, the M.S. degree from Osaka University in 2012, and the Ph.D. degree from Nagoya University in 2017. He is currently an Associate Professor in the School of Software Technology at Dalian University of Technology. His research interests include computer vision, multi-modal data fusion, and scene understanding.



Yi Wang received the B.E. and Ph.D. degrees in computer science and technology from Jilin University, China, in 2002 and 2009, respectively. Since 2009, she has been with the Dalian University of Technology, China. She is currently an Associate Professor. Her research interests include machine learning, image processing, and computer vision.